ASHWIN DHANASAMY

TECHNICAL SKILLS

Programming & Data: Python, SQL, Go, Java, PySpark, Pandas, NumPy, PostgreSQL, BigQuery
GenAI, LLMs & RAG: Embeddings, Semantic Search, Hybrid Retrieval (Dense + BM25), Reranking, pgvector, Pinecone
Agentic Systems & LLMOps: LangGraph, Multi-Agent Orchestration, Tool Use, MCP, Agent Workflows, Model Routing, Retries & Fallbacks, Human-in-the-Loop, LangSmith, RAGAS/DeepEval, LLM Tracing & Evaluation
ML: PyTorch, TensorFlow, Hugging Face, Scikit-learn, BERT Fine-tuning, LoRA/QLoRA, NER, LightGBM
Backend & Cloud Engineering: FastAPI, Spring Boot, REST APIs, Docker, Kubernetes, CI/CD, AWS (Bedrock, Lambda, Bedrock, EKS), GCP (Vertex AI, EKS, BigQuery, DataProc), Redis, Neo4j

PROFESSIONAL EXPERIENCE

Technical Co-Founder — Vybd (New York, NY) October 2023 – Present

- Architected a **multi-tenant LLM platform** using **FastAPI** and **LangGraph**, integrating **Neo4j, PostgreSQL, pgvector, Redis**; implemented **300+ automated tests** to validate system behavior and support reliable multi-client deployment.
- Reduced customs review effort and client tariff costs by **15%** by building a **document intelligence and RAG pipeline** combining OCR, fine-tuned **BERT** entity extraction, and LLM-based classification grounded in **17,000+ tariff codes**.
- Reduced research cost per session from **$4 to $1** by optimizing **hybrid retrieval** (dense embeddings + BM25) and reranking, while routing extraction tasks to smaller fine-tuned models and synthesis to frontier LLMs.
- Built and integrated **LangGraph agents** with **Odoo ERP** to automate purchase-order generation and operations reporting, implementing multi-step workflows to eliminate **20 hours/week** of manual work.
- Delivered AI solutions for **7 clients across 5 countries**, translating executive business challenges into deployed systems. For an apparel brand, reduced excess inventory by **28%** and shipping costs by **22%**, recovering **$180K/year**.
- Prepared a training-ready dataset of **1000 consumer brands** from agent research by deduplicating entities, normalizing schemas, dropping uncited claims, and labeling growth outcomes with **time-based splits** to prevent leakage (**BigQuery**).
- Deployed LLM and agentic systems on **Vertex AI** with **enterprise integrations, robust agent tracing, automated LLM evaluations, and failure handling** to monitor reliability, retrieval quality, and output correctness.
- **Optimized LLM inference cost and performance** through **model routing, prompt and context optimization, token usage reduction, and latency-aware model selection**, balancing response quality and throughput.

Software Engineer — Nichesolv (Bengaluru, India) October 2020 – December 2022

- Built and shipped backend services using **Java, Spring Boot, PostgreSQL, and Elasticsearch**, integrating enterprise data sources through REST APIs across social media, sports analytics, and education products; automated testing and delivery with **Docker and Jenkins CI/CD**.
- Owned client-facing requirements gathering and sprint planning across **3 product lines**, translating business needs into technical specifications, delivery roadmaps, and cross-functional execution plans.
- Shaped a **computer vision tennis analytics product** as its **product manager and subject-matter expert**, turning coach and player needs into model requirements and a delivery roadmap for the engineering team.
- Took **VGG16** and **YOLO** models from raw match video to production by building labeled **training datasets** (frame extraction, annotation), **deploying the models on AWS**, and integrating them into the product's **Spring Boot** backend.

OPEN SOURCE

Perses MCP Server (CNCF observability project): built Go-based MCP tools that let LLM agents manage dashboards, datasources, and variables. [code]
Vito: built a prospecting tool that finds, qualifies and researches companies, then drafts sourced outreach. [code]

EDUCATION

Stevens Institute of Technology — MS in Machine Learning May 2024
Anna University — BTech in Information Technology September 2020

LEADERSHIP & ACHIEVEMENTS

- Managed a **$400K annual sports budget** as the elected student representative for all university athletics, and raised **$50K in sponsorship** to run the university's sports festival.
- Tennis: Ranked **#1 in the state** for multiple years, and reached the **top 50 in the national men's rankings at age 15**.